Geometric probability

L3-X3 Level 3 Senior · Combinatorics · Beyond the textbook

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Probability as length, area or volume, covering meeting problems, the broken stick and the triangle picture, gaps between random points, what "at random" must mean, and Buffon's needle.
By the end you can turn independent uniform choices into a point of a square or cube and find probabilities as areas or volumes; use the triangle picture and the symmetry of the gaps between random points to find probabilities and expected values; say which quantity is uniform in a problem, avoid Bertrand's trap, and use Buffon's needle and Barbier's theorem.
Prerequisites: Probability by counting (L2-CO2), Expected value and probability tools (L3-CO2).
Contests: SMC, Senior Kangaroo, AMC 12, AIME. **Time:** about 90 minutes.

1 Lesson

Ana chooses a real number a uniformly at random from $[0,1]$, and Ben independently chooses b uniformly at random from $[0,2]$. What is the probability that $b > 3a$? There is no list of equally likely outcomes to count. Instead we regard the pair (a,b) as one point of the rectangle $[0,1]\times[0,2]$, of area 2. The event $b > 3a$ is the triangle with vertices $(0,0)$, $(0,2)$ and $(\frac{2}{3},2)$, of area $\frac{2}{3}$, so the probability is $\frac{2/3}{2} = \frac{1}{3}$.

This is *geometric probability*, in which probabilities are ratios of lengths, areas or volumes. L2-CO2 used the idea for one point in a square. The subject begins with Buffon's needle problem (posed in 1733, solved in his *Essai d'arithmétique morale* of 1777), and the paradox in Section 5 is Bertrand's (*Calcul des probabilités*, 1889). School courses stop at "shade the region and find its area". Contest problems also need the right choice of space, a change of variables that makes the region symmetric, and a clear idea of what "at random" means. We assume L2-CO2 and L3-CO2 (linearity of expectation).

1.1 Measure instead of counting

Principle.
If a point is chosen uniformly at random from a region R of finite positive length, area or volume $|R|$, then for every region $E \subseteq R$,

$$\mathbb{P}(E) = \frac{|E|}{|R|}.$$

If $x_1,\dots,x_n$ are chosen independently and uniformly from intervals $I_1,\dots,I_n$, then $(x_1,\dots,x_n)$ is a uniform point of the box $I_1\times\cdots\times I_n$.

For $n = 2$ the second statement holds for every rectangle $[p,q]\times[r,s]$ inside the box, since independence gives probability $\frac{q-p}{|I_1|}\cdot\frac{s-r}{|I_2|}$, which is the ratio of areas, and every region we meet is built from rectangles by cutting and taking limits. Three consequences are worth remembering. A line or curve has zero area, so $<$ and $\le$ give the same answer and "the two numbers are equal"

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has probability 0. A map that multiplies every area by the same factor (a translation, rotation, reflection, shear or scaling) takes a uniform point to a uniform point. And if the region and the event are both unchanged by a symmetry, we may work in one piece of the region cut out by the symmetries. For two random points on a circle, rotate so that the first is fixed, and the problem has one variable instead of two.

1.2 Two random times

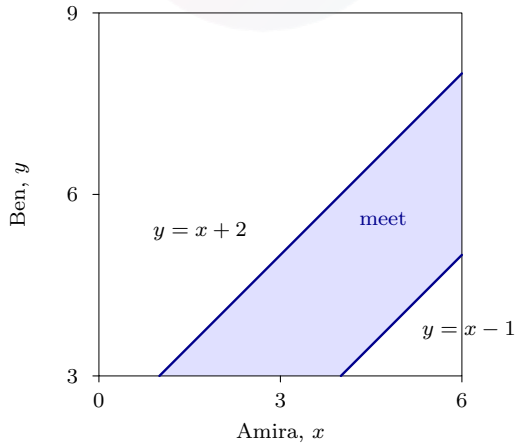
Two people arriving at random times and waiting for each other is the standard first application. Put one time on each axis. "They meet" becomes $-c \le y - x \le d$, a band between two lines of slope 1, and its complement is made of triangles and other simple polygons.

Example 1.
Amira arrives at a café at a uniformly random time between 12:00 and 13:00 and waits 20 minutes for Ben before leaving. Ben arrives, independently, at a uniformly random time between 12:30 and 13:30 and waits 10 minutes. What is the probability that they meet?

Solution.
Measure time in units of ten minutes after 12:00. Amira arrives at $x \in [0,6]$ and Ben at $y \in [3,9]$, so (x,y) is a uniform point of a square of area 36. They meet if Ben arrives while Amira is waiting, $x \le y \le x+2$, or Amira arrives while Ben is waiting, $y \le x \le y+1$. Together these say $-1 \le y - x \le 2$.
The points above the line $y = x+2$ (Ben arrives too late) form the square minus the triangle with vertices $(1,3)$, $(6,3)$, $(6,8)$, so they have area $36 - \frac{25}{2} = \frac{47}{2}$. The points below $y = x-1$ (Amira arrives too late) form the triangle $(4,3)$, $(6,3)$, $(6,5)$ of area 2. Hence the probability is

$$\frac{36 - \frac{47}{2} - 2}{36} = \frac{21/2}{36} = \frac{7}{24}.$$

The waits differ and the windows overlap only in part, so no remembered formula applies. Write the condition as an inequality on $y - x$ before drawing anything.

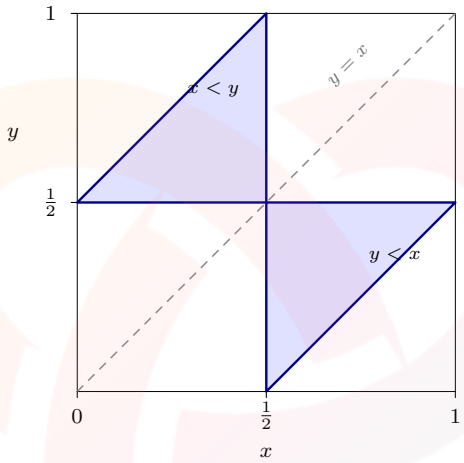


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1.3 The broken stick

Theorem 1 (broken stick).
A stick of length 1 is broken at two points chosen independently and uniformly at random. The three pieces form a triangle with probability $\frac{1}{4}$.

Proof. Three positive lengths with sum 1 form a non-degenerate triangle if and only if each is less than the sum of the other two, that is, each is less than $\frac{1}{2}$. Let the break points be at x and y. The event is unchanged by swapping x and y, so its probability is the same given $x < y$ as given $y < x$, and we may assume $x < y$, giving the triangle $0 < x < y < 1$ of area $\frac{1}{2}$. The pieces are x, $y - x$, $1 - y$, and the conditions $x < \frac{1}{2}$, $y > \frac{1}{2}$, $y - x < \frac{1}{2}$ describe the triangle with vertices $(0, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, 1)$, of area $\frac{1}{8}$. The probability is $\frac{1/8}{1/2} = \frac{1}{4}$. ■



The square handles this case, but for less symmetric conditions on x, $y - x$, $1 - y$ the square produces polygons with awkward vertices. The next picture treats the three pieces alike.

Theorem 2 (the triangle picture).
Let T be an equilateral triangle of height 1. The distances a, b, c from a point of T to the three sides satisfy $a + b + c = 1$ (Viviani), and every triple of positive numbers with sum 1 comes from exactly one point. If a stick of length 1 is broken as in Theorem 1, the three pieces have the distribution of (a, b, c) for a uniform random point of T. Moreover, for $p, q, r \geq 0$ with $p + q + r < 1$, the points with $a \geq p$, $b \geq q$, $c \geq r$ form an equilateral triangle of height $1 - p - q - r$.

Proof. Joining a point of T to the vertices cuts T into three triangles whose bases are the sides, of length s say, so $\frac{1}{2}s(a + b + c) = [T] = \frac{1}{2}s \cdot 1$. Given $x < y$, the point (x, y) is uniform on the triangle $0 < x < y < 1$, and the shear $(x, y) \mapsto (x, y - x)$ makes (a, b) uniform on the triangle $a, b > 0$, $a + b < 1$. The point of T with distances a and b to two given sides is an affine function of (a, b), which multiplies all areas by the same factor, so the point is uniform in T. The case $y < x$ is the same. For the last statement, the three lines $a = p$, $b = q$, $c = r$ are parallel to the sides of T, so they bound a triangle similar to T, and on its side $a = p$ we have $b + c = 1 - p$, and at its opposite vertex $b = q$, $c = r$, $a = 1 - q - r$, so its height is $1 - p - q - r$. ■

The condition of Theorem 1 removes from T the three corners $a \geq \frac{1}{2}$, $b \geq \frac{1}{2}$, $c \geq \frac{1}{2}$, each of height $\frac{1}{2}$ and area $\frac{1}{4}[T]$, which leaves the medial triangle and the answer $\frac{1}{4}$ again.

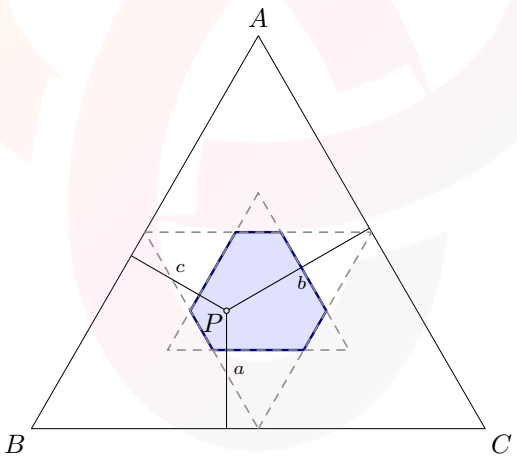
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Example 2.
A stick of length 1 is broken at two points chosen independently and uniformly at random. What is the probability that every piece is at least $\frac{1}{5}$ long and the three pieces form a triangle?

Solution.
First in the square, with $x < y$. The conditions are $x \geq \frac{1}{5}$, $y - x \geq \frac{1}{5}$, $1 - y \geq \frac{1}{5}$, $x < \frac{1}{2}$, $y - x < \frac{1}{2}$ and $y > \frac{1}{2}$, which are six lines, and the region is a hexagon whose six vertices must each be found as an intersection before the shoelace formula gives its area. This can be done, with several chances to slip.
In the triangle picture all six conditions have the same form. By Theorem 2, $a, b, c \geq \frac{1}{5}$ is an equilateral triangle of height $1 - \frac{3}{5} = \frac{2}{5}$, with area $\frac{4}{25}[T]$. From this triangle we remove the points with some piece at least $\frac{1}{2}$. The points with $a \geq \frac{1}{2}$, $b \geq \frac{1}{5}$, $c \geq \frac{1}{5}$ form a triangle of height $1 - \frac{1}{2} - \frac{2}{5} = \frac{1}{10}$ and area $\frac{1}{100}[T]$, and the same holds for b and for c. No two of these corners overlap, since two pieces cannot both be at least $\frac{1}{2}$ except with probability 0. The probability is

$$\frac{4}{25} - \frac{3}{100} = \frac{13}{100}.$$

In the triangle picture the hexagon is a triangle with three corners cut off, so its area needs no coordinates.



1.4 Gaps between random points

Theorem 3 (gaps).
Choose n points independently and uniformly from $[0, 1]$. They cut the interval into $n + 1$ gaps $g_1, \ldots, g_{n+1}$, from left to right. Every permutation of the gaps has the same distribution as $(g_1, \ldots, g_{n+1})$. In particular $\mathbb{E}[g_i] = \frac{1}{n+1}$ for each i, and the kth smallest point has expected value $\frac{k}{n+1}$.

Proof. For $n = 2$ this is Theorem 2, since permuting a, b, c is a symmetry of T. In general let $u_1 < \cdots < u_n$ be the points in increasing order. The cube $[0, 1]^n$ is cut into $n!$ congruent pieces, one for each ordering, so $(u_1, \ldots, u_n)$ is uniform on the piece $0 < u_1 < \cdots < u_n < 1$. The map to $(g_1, \ldots, g_n) = (u_1, u_2 - u_1, \ldots, u_n - u_{n-1})$ is a shear, so $(g_1, \ldots, g_n)$ is uniform on the region $g_i > 0$, $g_1 + \cdots + g_n < 1$, with $g_{n+1} = 1 - g_1 - \cdots - g_n$. Exchanging any two of the $n + 1$ gaps maps this region to itself by an affine map of determinant ± 1, so the map keeps the uniform distribution.

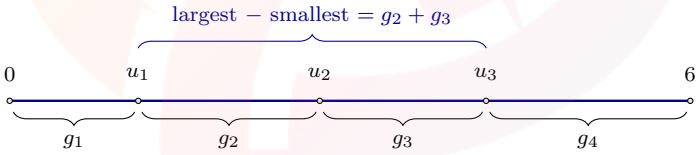
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The gaps have equal expectations and sum 1, so each has expectation $\frac{1}{n+1}$, and $u_k = g_1 + \cdots + g_k$. ■

The same fact holds on a circle, where $n+1$ random points on a circle of circumference 1, cut at one of them, give n random points on $[0,1]$. Theorem 3 replaces an integral over n variables by one line of symmetry.

Example 3.
Three numbers are chosen independently and uniformly at random from $[0,6]$. Find the expected value of the largest minus the smallest.

Solution.
First the standard way, for two numbers x, y in $[0,1]$. The expected value of $|x-y|$ is the volume under the surface $z = |x-y|$ above the unit square. Over the half $y < x$ the solid $0 \le z \le x - y$ is the tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(1,1,0)$, $(1,0,1)$, since its top face lies in the plane $z = x - y$ through three of them. Its base in the plane $x = 1$ has area $\frac{1}{2}$ and the apex is at distance 1, so its volume is $\frac{1}{6}$, and $\mathbb{E}|x-y| = \frac{1}{3}$. For three numbers the same method needs a volume in four dimensions, or six integrals, one for each ordering.
With Theorem 3, scale $[0,6]$ to $[0,1]$. Three points make four gaps, each with expected length $\frac{1}{4}$, and the largest minus the smallest is $g_2 + g_3$, with expected value $\frac{1}{2}$. On $[0,6]$ the answer is 3. The same argument recovers the case of two points, since $|x-y|$ is the middle one of three gaps, so its expected value is $\frac{1}{3}$.



1.5 What "at random" must mean

Bertrand asked for the probability that a chord of a circle of radius 1, chosen at random, is longer than $\sqrt{3}$, the side of the inscribed equilateral triangle. A chord at distance d from the centre has length $2\sqrt{1-d^2}$, which is more than $\sqrt{3}$ if and only if $d < \frac{1}{2}$. Here are three readings of "at random".

1. The endpoints are independent and uniform on the circle. Rotate so that the first endpoint is fixed. The chord is long when the second endpoint lies on the arc of 120° opposite the first, so the probability is $\frac{1}{3}$.
2. A radius is chosen, then a point uniformly on that radius, and the chord is perpendicular to the radius at that point. Then d is uniform on $[0,1]$ and the probability is $\frac{1}{2}$.
3. The midpoint of the chord is uniform in the disc. The chord is long when the midpoint lies in the disc of radius $\frac{1}{2}$, so the probability is $\frac{1}{4}$.

None of these is wrong. They answer three different questions, because "at random" has no meaning until the statement says which quantity is uniform. Contest statements do say ("the endpoints are chosen independently and uniformly on the circle"), and the step that loses marks is translating the statement into a region in which a different quantity is uniform. Two facts come up often. The distance d from the centre of a uniform point of a disc of radius R is not uniform,

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since $\mathbb{P}(d \le r) = \frac{r^2}{R^2}$, so d^2 is the quantity that is uniform on $[0, R^2]$, and a uniform angle θ does not make $\sin\theta$ uniform.

1.6 Buffon's needle

Theorem 4 (Buffon, 1777).
A floor is ruled with parallel lines at distance d apart, and a needle of length $\ell \le d$ is dropped at random. The distance from its centre to the line below is uniform on $[0, d)$, its angle with the lines is uniform, and the two are independent. The needle crosses a line with probability $\frac{2\ell}{\pi d}$.

First proof. Let y be the distance from the centre to the nearest line, uniform on $[0, \frac{d}{2}]$, and let θ be the acute angle between the needle and the lines, uniform on $[0, \frac{\pi}{2}]$. The needle reaches the line if and only if $y \le \frac{\ell}{2}\sin\theta$. The point (θ, y) is uniform in a rectangle of area $\frac{\pi d}{4}$, and the region under the curve $y = \frac{\ell}{2}\sin\theta$ has area $\frac{\ell}{2}\int_0^{\pi/2}\sin\theta\,d\theta = \frac{\ell}{2}$ (this is the one step that needs calculus, and the second proof avoids this). The ratio is $\frac{2\ell}{\pi d}$. ■

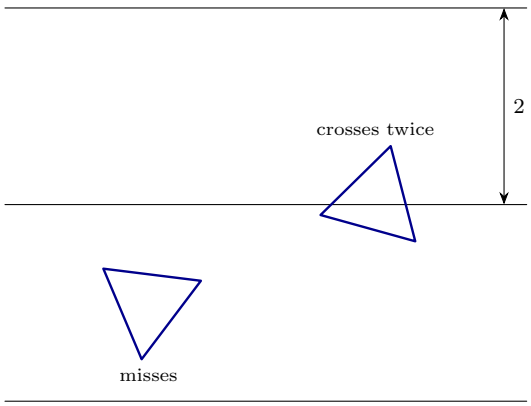
Second proof (Barbier, 1860). Let N be the number of crossings and $f(\ell) = \mathbb{E}[N]$ for a needle of length ℓ, now of any length. A needle of length $\ell_1 + \ell_2$ is two needles of lengths ℓ_1 and ℓ_2 joined end to end, and each of them on its own is dropped at random, so linearity of expectation gives $f(\ell_1 + \ell_2) = f(\ell_1) + f(\ell_2)$. Hence $f(\ell) = c\ell$ for rational multiples of a fixed length, and since f is increasing, for all ℓ. The same argument applies to a rigid wire made of straight pieces of total length L, giving $\mathbb{E}[N] = cL$, and by approximation to a curved wire. A circle of diameter d meets the lines in exactly two points (except with probability 0) and has length πd, so $c\pi d = 2$. Finally a needle with $\ell \le d$ crosses at most once, except with probability 0, so the probability of a crossing equals $\mathbb{E}[N] = \frac{2\ell}{\pi d}$. ■

The second proof gives more than the theorem. For any rigid wire of length L, dropped with a uniform random offset and a uniform random rotation, the expected number of crossings is $\frac{2L}{\pi d}$. This is known as Buffon's noodle.

Example 4.
A wire bent into an equilateral triangle of side 1 is dropped at random onto a floor ruled with parallel lines 2 apart (uniform random position and uniform random rotation, as in Theorem 4). What is the probability that the wire meets a line?

Solution.
Working with Theorem 4 directly, we would need, for each angle, the width of the triangle perpendicular to the lines, and then an average of that width over the angle. Instead use Barbier's result. The wire has length 3, so the expected number of crossings is $\frac{2\cdot 3}{2\pi} = \frac{3}{\pi}$.
Any two points of the triangle are at most $1 < 2$ apart, so the wire meets at most one line, and a line that meets a triangle, and does not pass through a vertex, crosses its boundary exactly twice. So N is 0 or 2 with probability 1, and $\mathbb{P}(N = 2) = \frac{1}{2}\mathbb{E}[N] = \frac{3}{2\pi} \approx 0.48$.
A written solution must state both facts used, namely that the wire is convex and that the wire is too small to reach two lines.

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Common traps.

- Making the wrong quantity uniform, as in Bertrand's paradox. Before drawing a region, check that its coordinates are the quantities the statement makes uniform.
- Using the wrong region. The centre of a coin of radius r placed inside a rectangle ranges over a smaller rectangle, and that is the area to divide by.
- Assuming $x < y$ by symmetry and then dividing by the area of the whole square instead of half of the square.
- Changing the experiment. Breaking a stick at one random point and then breaking the longer piece at a random point is a different experiment from Theorem 1, and the probability of a triangle is no longer $\frac{1}{4}$.
- Writing $\mathbb{P} = \mathbb{E}[N]$ in a Buffon problem when N can be 2 or more.
- In a written solution, a marker needs the space and the reason the point is uniform, the region with its vertices, and the area computation. "By the diagram" is not enough.

1.7 How to spot this in a contest

1. The words "uniformly at random" or "independently" attached to real numbers, times or points, rather than to a finite list.
2. Two arrival times and waiting times, so draw the square and the band $-c \leq y - x \leq d$.
3. Three numbers suggest a cube, and the volume of a corner or a slab. Three quantities with a fixed sum suggest the triangle picture of Theorem 2.
4. A segment or a circle cut at random points, or the expected value of the largest, smallest or kth point suggest Theorem 3.
5. A random chord, line or direction, so decide what is uniform before anything else. Parallel lines and a dropped object suggest Buffon, and for any wire, Barbier.
6. A symmetric region, so cut the region into congruent pieces by its symmetries and work in one piece.

2 Practice problems

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The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ A train arrives at a station at a uniformly random time between 9:00 and 9:20 and waits there for 4 minutes before leaving. Lena arrives at the station, independently, at a uniformly random time between 9:10 and 9:30. What is the probability that she catches the train?

Problem 2 ⋆⋆ A point P is chosen uniformly at random inside the triangle with vertices $(0,0)$, $(4,0)$ and $(0,3)$. What is the probability that P is farther from the hypotenuse than from each of the two legs?

Problem 3 ⋆⋆ A point P is chosen uniformly at random inside a circle of radius 2 with centre O. The chord of the circle through P perpendicular to OP has length L. Find the expected value of L^2.

Problem 4 ⋆⋆⋆ A stick of length 1 is broken at two points chosen independently and uniformly at random. What is the probability that the longest of the three pieces is at most twice as long as the shortest?

Problem 5 ⋆⋆⋆ Three friends arrive at a café at independent, uniformly random times between 18:00 and 19:00. Each stays for exactly 15 minutes and then leaves. What is the probability that at some moment all three are in the café together?

Problem 6 ⋆⋆⋆⋆ A wire bent into a square of side 1 is dropped at random onto a floor ruled with parallel lines 1 apart (uniform random position and uniform random rotation). Find the expected number of lines that the square meets.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME II 2010 Problem 2** ⋆
   The points at distance at least r from every side of a square form a smaller square.
2. **AMC 10A 2012 Problem 25** ⋆⋆
   Three numbers are a point of a cube. Sort them, then remove the forced gaps to get a smaller cube.
3. **HMMT November 2023 Guts Round Problem 10** ⋆⋆
   One variable, so write the three triangle inequalities and measure the interval where all hold.
4. **AIME 1998 Problem 9** ⋆⋆
   A square with a band about its diagonal, and the complement is two triangles.
5. **AMC 12A 2015 Problem 23** ⋆⋆⋆
   By symmetry fix the side and half of that side holding the first point, then split by the side holding the second.
6. **AIME I 2017 Problem 8** ⋆⋆⋆
   Q and R lie on the circle with diameter OP. Find QR in terms of a and b, and the condition becomes a band in a square.

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7. **AIME I 2004 Problem 10** ⋆⋆⋆
   The centre ranges over a smaller rectangle. Where must the centre be to stay clear of the diagonal?

8. **HMMT November 2020 Guts Round Problem 23** ⋆⋆⋆
   The overlap of the two squares is a rectangle whose sides depend on the two coordinate differences separately.

9. **HMMT November 2023 Guts Round Problem 33** ⋆⋆⋆
   Write XY with vectors from the two centres, and the cross terms have expected value zero by symmetry.

10. **AIME II 2023 Problem 6** ⋆⋆⋆
    Split by the squares containing A and B. A rectangle made of two squares is convex, which disposes of most cases.

11. **HMMT February 2020 Combinatorics Problem 8** ⋆⋆⋆⋆
    A side misses the inner circle exactly when its arc is short. Use the gaps between the four points.

12. **HMMT February 2021 Guts Round Problem 24** ⋆⋆⋆⋆
    The cross-section depends only on $x + y + z$. Find the perimeter as a function of that sum, then average over its distribution.

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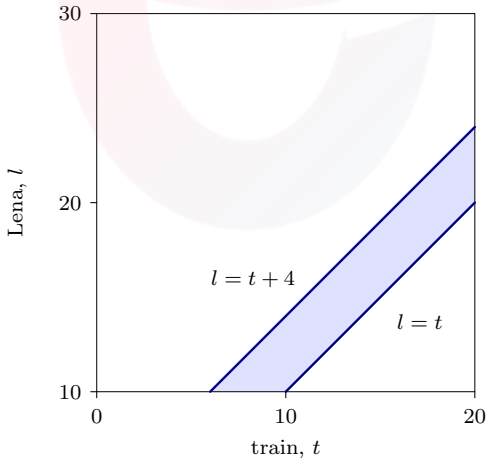
4 Hints

1. Put the train's arrival on one axis and Lena's on the other. Write the condition as an inequality on the difference of the two times.
2. The points equidistant from two sides of a triangle lie on the bisector of the angle between them. Which two bisectors bound the region, and where do they meet?
3. Write L^2 in terms of $d = OP$. Which function of d is uniformly distributed?
4. By Theorem 2 the pieces (a, b, c) are symmetric. Assume $a \le b \le c$ and work in the (a, b) plane with $c = 1 - a - b$.
5. All three are present at some moment exactly when the last to arrive comes before the first to leave. Sort the three times and slice the region by the earliest time.
6. Barbier gives the expected number of crossings. How many times does a line that meets the square cross the square?

5 Solutions

Problem 1

A train arrives at a station at a uniformly random time between 9:00 and 9:20 and waits there for 4 minutes before leaving. Lena arrives at the station, independently, at a uniformly random time between 9:10 and 9:30. What is the probability that she catches the train?



Let the train arrive t minutes after 9:00 and Lena l minutes after 9:00, so (t, l) is a uniform point of the square $[0, 20] \times [10, 30]$, of area 400. She catches the train if and only if $t \le l \le t + 4$. Since $t + 4 \le 24 < 30$, the top edge of the square does not cut this strip, and $l \ge 10$ forces $t \ge 6$.

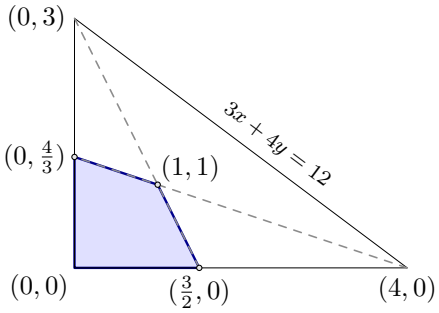
The part of the strip in the square is the quadrilateral with vertices $(6, 10)$, $(10, 10)$, $(20, 20)$, $(20, 24)$. The region consists of the parallelogram $(10, 10)$, $(20, 20)$, $(20, 24)$, $(10, 14)$, of area $10 \cdot 4 = 40$, and the triangle $(6, 10)$, $(10, 10)$, $(10, 14)$, of area 8. The probability is $\frac{48}{400} = \frac{3}{25}$.

Answer: $\frac{3}{25}$

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Problem 2

A point P is chosen uniformly at random inside the triangle with vertices $(0,0)$, $(4,0)$ and $(0,3)$. What is the probability that P is farther from the hypotenuse than from each of the two legs?



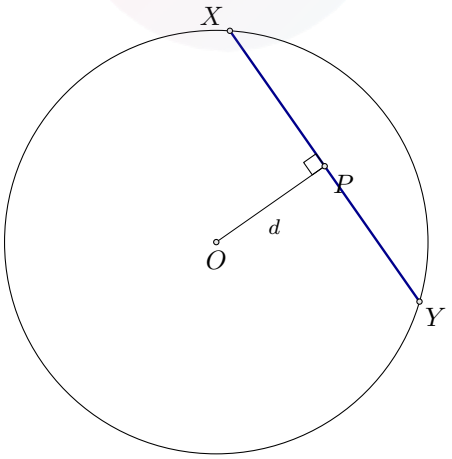
For $P = (x, y)$ inside the triangle the distances to the legs are x and y, and the distance to the hypotenuse $3x + 4y = 12$ is $h = \frac{12-3x-4y}{5}$. The condition $h > x$ reads $2x + y < 3$, and $h > y$ reads $x + 3y < 4$. These lines are the bisectors of the angles at $(0,3)$ and at $(4,0)$, as they must be, and they meet at $(1,1)$, the incentre (the inradius is $\frac{3+4-5}{2} = 1$).

So the event is the quadrilateral with vertices $(0,0)$, $(\frac{3}{2},0)$, $(1,1)$, $(0,\frac{4}{3})$, where the two bisectors meet the legs. Joining $(0,0)$ to $(1,1)$ cuts the quadrilateral into two triangles with bases $\frac{3}{2}$ and $\frac{4}{3}$ on the axes and heights 1, so its area is $\frac{3}{4} + \frac{2}{3} = \frac{17}{12}$. The whole triangle has area 6, so the probability is $\frac{17/12}{6} = \frac{17}{72}$.

Answer: $\frac{17}{72}$

Problem 3

A point P is chosen uniformly at random inside a circle of radius 2 with centre O. The chord of the circle through P perpendicular to OP has length L. Find the expected value of L^2.



Let $d = OP$. The chord has P as its midpoint, so by Pythagoras $\left(\frac{L}{2}\right)^2 = 4 - d^2$ and $L^2 = 4(4 - d^2)$. For $0 \le s \le 4$, the event $d^2 \le s$ means that P lies in the disc of radius $\sqrt{s}$, which has probability $\frac{\pi s}{4\pi} = \frac{s}{4}$. So d^2 is uniform on $[0,4]$ and $\mathbb{E}[d^2] = 2$. By linearity, $\mathbb{E}[L^2] = 4(4-2) = 8$.

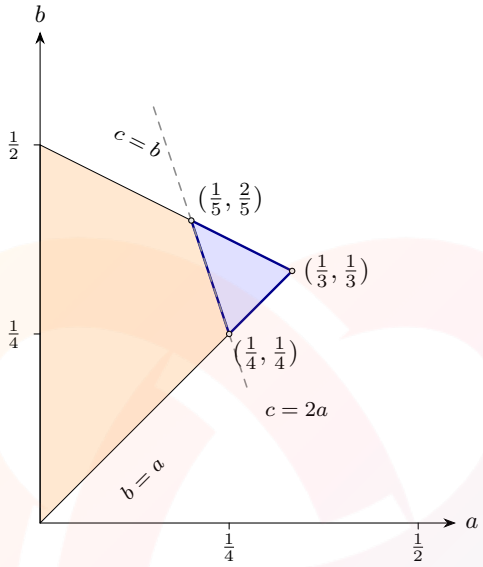
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Taking d itself to be uniform on $[0,2]$ would give $\mathbb{E}[d^2] = \frac{4}{3}$ and the wrong answer $\frac{32}{3}$, and this is the second reading of Bertrand's problem in place of the third.

Answer: 8

Problem 4

A stick of length 1 *is broken at two points chosen independently and uniformly at random. What is the probability that the longest of the three pieces is at most twice as long as the shortest?*



By Theorem 2 the pieces (a,b,c) have a distribution unchanged by permutations, and the event is symmetric in a, b, c. The six orderings are equally likely, so we may assume $a \le b \le c$ and compute the conditional probability. In the proof of Theorem 2, (a,b) is uniform on the triangle $a,b>0$, $a+b<1$, with $c = 1-a-b$.

The conditions $a \le b$ and $b \le c$, that is $b \ge a$ and $b \le \frac{1-a}{2}$, give the triangle with vertices $(0,0)$, $(\frac{1}{3},\frac{1}{3})$, $(0,\frac{1}{2})$, of area $\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{3} = \frac{1}{12}$. The extra condition $c \le 2a$ reads $b \ge 1-3a$. The line $b = 1-3a$ meets $b = a$ at $(\frac{1}{4},\frac{1}{4})$ and meets $b = \frac{1-a}{2}$ at $(\frac{1}{5},\frac{2}{5})$, so the event is the triangle with vertices $(\frac{1}{4},\frac{1}{4})$, $(\frac{1}{3},\frac{1}{3})$, $(\frac{1}{5},\frac{2}{5})$. By the shoelace formula its area is

$$\frac{1}{2}\left|\left(\tfrac{1}{3}-\tfrac{1}{4}\right)\left(\tfrac{2}{5}-\tfrac{1}{4}\right)-\left(\tfrac{1}{3}-\tfrac{1}{4}\right)\left(\tfrac{1}{5}-\tfrac{1}{4}\right)\right| = \frac{1}{2}\cdot\frac{1}{12}\cdot\frac{1}{5} = \frac{1}{120}.$$

The probability is $\frac{1/120}{1/12} = \frac{1}{10}$.

Answer: $\frac{1}{10}$

Problem 5

Three friends arrive at a café at independent, uniformly random times between 18:00 and 19:00. Each stays for exactly 15 *minutes and then leaves. What is the probability that at some moment all three are in the café together?*

Measure time in hours after 18:00, so the arrival times are independent and uniform on $[0,1]$, and let $t = \frac{1}{4}$. The intervals $[x, x+t]$, $[y, y+t]$, $[z, z+t]$ have a common point if and only if the latest arrival is at most the earliest departure, that is, the largest time minus the smallest is at most t.

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The six orderings of x, y, z are equally likely, so the probability is $6V$, where V is the volume of the set $0 \le x \le y \le z \le 1$ with $z - x \le t$. Slice by the value s of x. The slice is the set of (y,z) with $s \le y \le z \le \min(s+t,1)$, a right-angled isosceles triangle with legs $\min(t, 1-s)$. For $0 \le s \le 1-t$ the legs are t, so this part of the solid is a prism of cross-section $\frac{t^2}{2}$ and length $1-t$, with volume $\frac{t^2(1-t)}{2}$. For $1-t \le s \le 1$ the slices shrink linearly to a point, so this part is a pyramid with base area $\frac{t^2}{2}$ and height t, with volume $\frac{t^3}{6}$. Hence

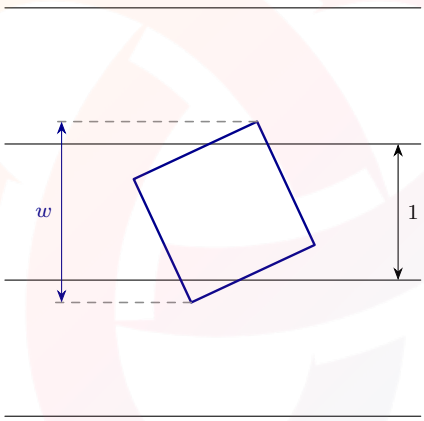
$$6V = 3t^2(1-t) + t^3 = 3t^2 - 2t^3 = \frac{3}{16} - \frac{2}{64} = \frac{5}{32}.$$

As a check, $t = 1$ gives 1 and $t = 0$ gives 0, as they should.

Answer: $\frac{5}{32}$

Problem 6

A wire bent into a square of side 1 *is dropped at random onto a floor ruled with parallel lines* 1 *apart (uniform random position and uniform random rotation). Find the expected number of lines that the square meets.*



The wire has length 4, so by Barbier's form of Theorem 4 the expected number of crossings is $\mathbb{E}[N] = \frac{2 \cdot 4}{\pi \cdot 1} = \frac{8}{\pi}$. Here a crossing count can be as large as 4, because the square is wider than the spacing in most directions, so $\mathbb{E}[N]$ is not a probability. But the square is convex, so every line that meets the square, other than a line through a vertex or along a side (which happens with probability 0), crosses its boundary exactly twice. Hence the number of lines met is $\frac{N}{2}$, and its expected value is $\frac{4}{\pi} \approx 1.27$.

A second derivation, which uses one integral, checks this. If a side makes the angle θ with the lines, the square's width perpendicular to the lines is $w = |\cos\theta| + |\sin\theta|$, and for a uniform random offset the expected number of lines met by a segment of length w perpendicular to them is w (lines 1 apart). The mean of $|\cos\theta|$ over a uniform angle is $\frac{2}{\pi}\int_0^{\pi/2} \cos\theta \, d\theta = \frac{2}{\pi}$, and the same for $|\sin\theta|$, so $\mathbb{E}[w] = \frac{4}{\pi}$. Note that $w \ge 1$ always, so the square meets at least one line with probability 1.

Answer: $\frac{4}{\pi}$

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